

Clearly describe the project topic, problem, and overall objective.

Topic

As the grid begins to rely further and further upon renewable resources, battery storage is going to be a lot more common in order to flatten the energy surplus during sunny and windy hours, as well as raise the energy deficit during cloudy/nighttime and calm hours. This will require more advanced battery storage algorithms in order to optimize energy sales while considering the day ahead and real-time prices.

Problem

Battery investors don't know whether it is more optimal to purchase energy a day in advance or wait to buy energy at the time it is needed.

Overall objective

Our overall objective is to create a prototype of a model that could be implemented and integrated into a battery facility to help the facility behave optimally by learning optimal conditions for charging and discharging batteries.

Anna Tolles has summarized the project topic, problem, and overall objectives.

Identify the main Data Science question(s) or problem(s) the project will address.

Problems

Are we able to create a model that will accurately predict real time price 24 hours in advance? How can predicted battery prices determine when a battery should be charged/discharged?

How we will address them

We will address this by testing multiple types of time series models using weather data, outage data, and day head market prices to see which model can most accurately

predict real time prices. To address the problem of determining the best time to charge and discharge a battery, we will use both mathematical algorithms and reinforcement learning to optimize profit of the battery entity.

Anna Tolles has identified data science questions and problems.

Identify and describe the dataset(s), tools, libraries, software, APIs, or other resources required for the project.

Datasets

We will be using the real-time market, day-ahead market, outage, load, solar, and wind tables from GridStatus. Each of us created accounts to get API keys to pull specific timeframes of data. We'll be using the Python library pandas for dataframe manipulation, matplotlib for data visualization, gymnasium for RL environment building and sklearn and tensorflow for model building and tuning. We are using Jira for project management and task tracking. We are using Github for workflow and project collaboration.

Every team member has been using these tools to evaluate, visualize, and model the data

Describe the proposed Data Science workflow, including major steps such as data collection, preprocessing, exploration, feature development, modeling, and evaluation, as applicable.

Data Collection

For data collection, we each created a GridStatus account to get API keys and pull data for free. We then pulled each table into parquet files.

Preprocessing

We started out with six different datasets. The real-time market and load data were recorded in five minute intervals. The wind, solar, and day-ahead markets record in

intervals of 60 minutes. Finally the outage data was recorded in 24 hour intervals. Of all the datasets, most had a different range of interval recordings. Some starting in 2021, others in 2022. The first step was finding a uniform range of dates that were recorded across all datasets. In the wind and solar datasets there were five different projected recordings, one for each day in the five days leading up to a designated interval. We only wanted the projection from the day before, so I filtered out the other four days. Then I combined the wind and solar tables into a renewable energy dataset. I then combined the renewable energy table with the day-market as they both recorded in hourly intervals.

Transitioning to the two datasets that record in five minute intervals (real-time market and load data) based on previous preprocessing, theoretically these datasets should have the same number of rows. They did not. The next step was investigating why. I found that there were some missing intervals in the real-time market dataset that were just not recorded, so I went through and inserted these missing intervals in with missing values. This brought the two datasets to the same size. I then merged the load data with the real-time market data.

Finally I took our new dataset of renewable energy combined with the day-ahead market and took the 60 minute intervals and broke them up into 12 identical five minute intervals to merge with the real-time market and load dataset.

Kaitlyn Klapatauskas has done the bulk of the combining and preprocessing of our data.

Exploration

The primary part of exploration of the data columns was done using pre-existing knowledge gained from industry working at MISO and ATC. Some of the biggest things to note:

Day-Ahead Market (DA): A forward commitment market where market participants (generators, utilities, traders) submit bids and offers for electricity to be delivered the following day. Prices are calculated per hour (or half-hour) based on forecasted demand, expected generation, and grid constraints.

Real-Time Market (RT): A spot balancing market that runs constantly (typically clearing every 5 or 15 minutes). It accounts for the actual, real-time physical differences between what was scheduled in the Day-Ahead market and what is actually generated or consumed.

Battery Degradation: Batteries degrade with each charge and discharge as well as over time. While under short term usage this is negligible, we want our algorithm to

perform well for the life of the battery which can be anywhere from 4000-7000 charge and discharge cycles and 5 to 20 years.

There is some skew to some of the predictor variables which is something that Kaitlyn determined during her exploratory data analysis. This has guided our team to decide to perform center and scaling as this will likely improve the accuracy of our predictive models. We also found that we have some outliers which may be worth dropping because they may decrease the predictive power of the bulk of the data.

Alex Krawza and Zack Sickmann did extensive data exploration and did an amazing job explaining the meaning of each column to the other members.

Feature development

We are exploring calculating reserve margin as a potential in the import columns and features section. This is derived from outage forecast and load forecast and requires knowledge of total generation. The formula for reserve margin is $RM = [(T_{\text{Total Generation}} - O_{\text{Outage Forecast}}) - L_{\text{Load Forecast}}] / L_{\text{Load Forecast}} \times 100$. This gives us an idea of how much flexibility/excess generation is available. We will have to explore if there is a relationship between low reserve margins and the sudden price spikes which our optimization and reinforcement learning models could take advantage of.

Alex has been looking into future feature development directions.

Modeling

For modeling, we are testing out different time series models including moving average, time gap, exponential smoothing, arima/sarima/sarimax models, and LSTM. We also think it may be worth exploring non-time series models beginning with parametric prediction models and possibly more such as KNN regression, random forest regression, and more.

Reinforcement learning modelling process comprises building the battery storage environment by building a custom gymnasium environment. A DQN network is then constructed with an input layer the same size as the state space, which is the two

schedules, RTM and DAM. It is then projected on a 32 dimensional feature space, passes through a ReLU layer to introduce non-linearity, a hidden layer of input and output size 32, another ReLU layer onto another 32 input layer which outputs to a space the size of the schedule multiplied by 5 for the amount of possible actions, the actions being buy-rtm, buy-dam, hold, sell-rtm, and sell-dam.

Anna Tolles and Alex Krawza have begun building the time series models, Tommy has been researching non-time series models, and Zack Sickmann and Kaitlyn Klapatauskas have been experimenting with reinforcement learning models.

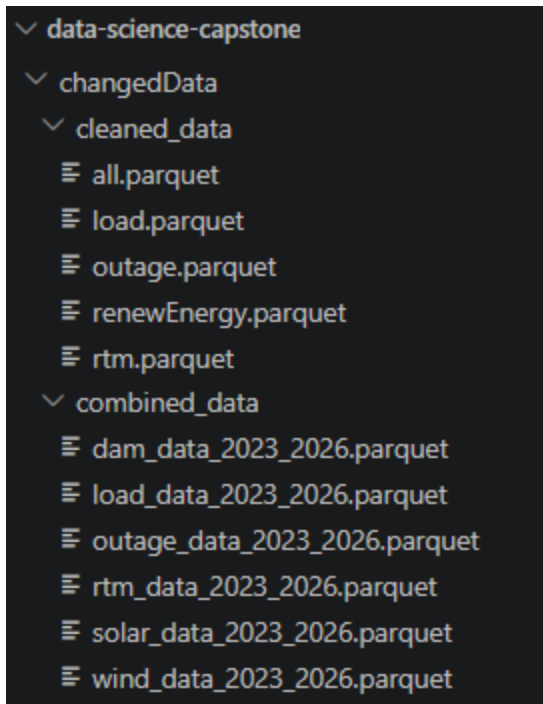
Evaluation

We will be evaluating model accuracy and performance by calculating the RMSE on a testing set and determining the model with the lowest RMSE. Special care will be taken to ensure that data leakage does not occur during this process, by splitting data into training and testing before model training.

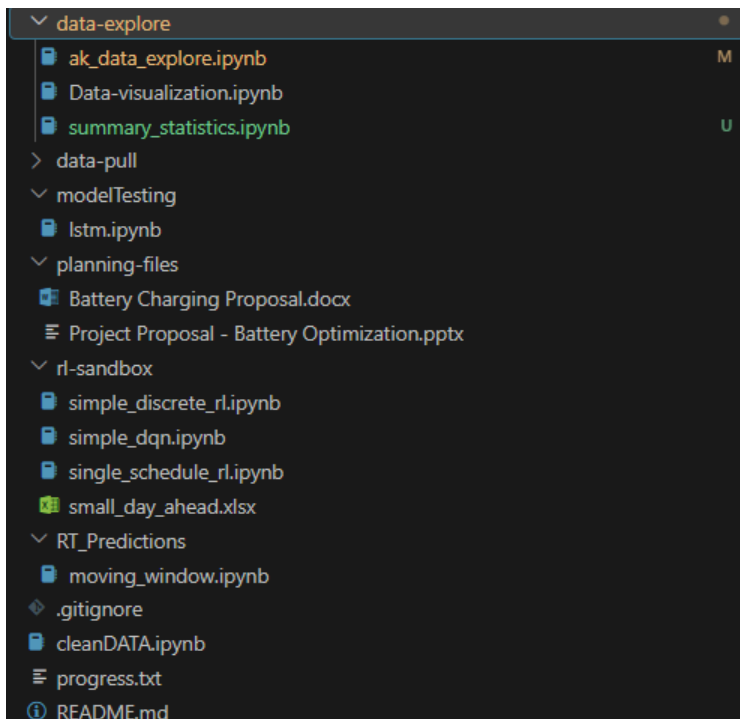
Evaluation of the RL model and mathematical based optimization algorithms is done by running the model over a testing set (likely going to be data from September to November 2026) and calculating the total profit the battery has accumulated over that time. This profit is then compared to baseline buy-sell algorithms and mathematically based optimization algorithms.

Zack Sickmann has determined how we will evaluate our model accuracy.

Document the current project setup and initial progress using screenshots, diagrams, examples, or other appropriate evidence.



We've pulled data from GridStatus, safely merged all of the tables, and cleaned it.



We've explored our data and have begun testing models and reinforcement learning on our tables.

Load the dataset(s) and perform an initial inspection.

```

#Function to combine data by file type
def combineDataFiles(folder, files, output_folder,
                    start_year=2023, end_year=2026):

    output_folder = Path(output_folder)
    output_folder.mkdir(parents=True, exist_ok=True)

    for file in files:

        dfs = []

        for year in range(start_year, end_year + 1):

            year_folder = Path(folder) / str(year)

            matching_files = year_folder.glob(
                f"{file}_data_*.parquet"
            )

            for file_path in matching_files:

                print(f"Loading: {file_path}")

                df = pd.read_parquet(file_path)

                df["year"] = year

                dfs.append(df)

        if dfs:

            combined_df = pd.concat(
                dfs,
                ignore_index=True
            )

            combined_df = combined_df.sort_values(
                "interval_end_utc"
            )

            output_path = (
                output_folder /
                f"{file}_data_{start_year}_{end_year}.parquet"
            )

            combined_df.to_parquet(
                output_path,
                index=False
            )

            print(
                f"Saved {len(combined_df):,} rows to {output_path}"
            )

        else:
            print(f"No files found for {file}")

```

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We loaded all the data from its individual files and combined them by dataset type across years. We also used the data wrangler tool in Visual Studio Code to inspect the data.

Alex Krawze did a lot of the initial download of the parquet files, while Kaitlyn Klapatauskas did the initial inspection.

Examine dataset dimensions, columns/features, data types, and representative observations.

For our initial investigation, Kaitlyn created a summary sheet of all of our datasets. Including shape, missingness, datatypes, and a five number summary of numerical features. This summary sheet is in our github under the data-pull folder labeled datasummary.pdf.

Kaitlyn Klapatauskas did the initial inspection of the data.

Check missing values, duplicates, unusual values, ranges, units, and potential data-quality issues, as applicable.

Upon initial investigation we found that there were intervals that were just skipped in the recordings. One of our first steps was to insert these skipped intervals as missing data. Besides this issue we found that our data was not that difficult to work with. We do have some extreme outliers so our next steps will be to deal with those outliers and skewness in our data.

Kaitlyn Klapatauskas did the initial inspection of the data.

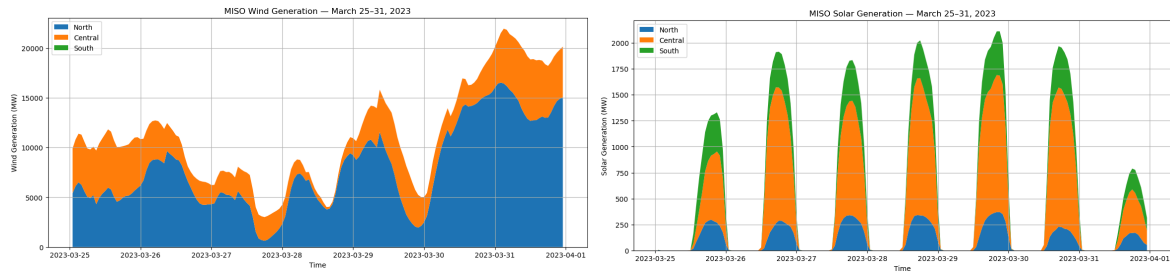
Generate appropriate summary statistics for important variables.

LMP price in the Day Ahead Market table and the Real Time Market table will be important in predicting real time lmp price 24 hours in advance. For LMP price in the day ahead table, it has a mean price of 34.69 dollars, with a standard deviation of 32.51 dollars. For LMP price in the real time table, it has a mean price of 34.41 dollars, and a standard deviation of 83.79 dollars. For the wind and solar tables, the MISO columns will be important to predict prices based on forecasted generated energy. For miso in the wind table, it has a mean generated energy of 11499.31 watts, with a standard deviation of 6084.82 watts. For miso in the solar table, it has a mean generated energy of 2418.38 watts, with a standard deviation of 3935.49 watts.

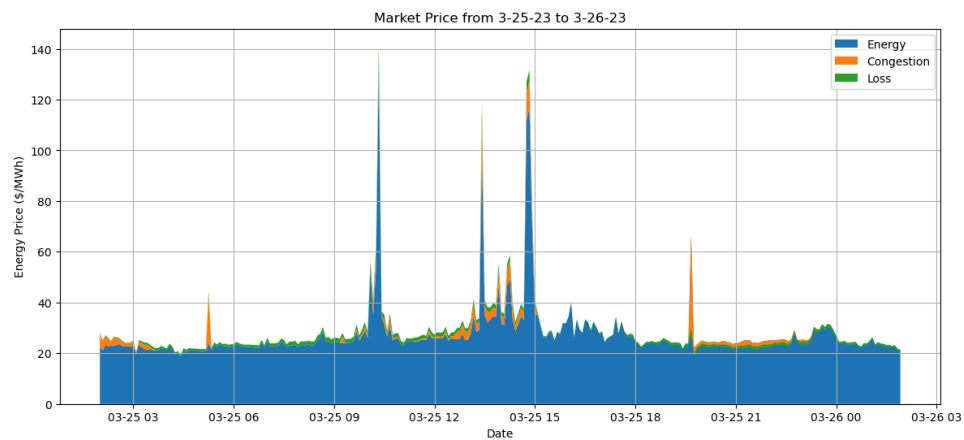
Anna Tolles has calculated the summary statistics for the above variables.

Create meaningful visualizations to explore important variables, distributions, relationships, patterns, or trends.

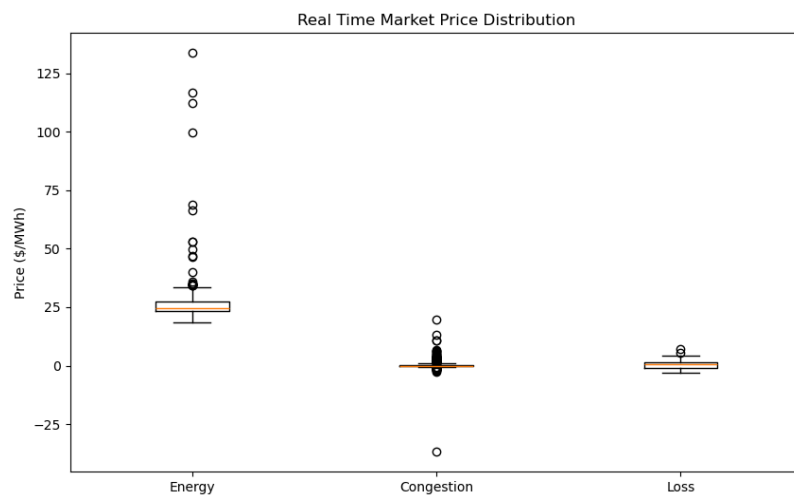
Data Visualization



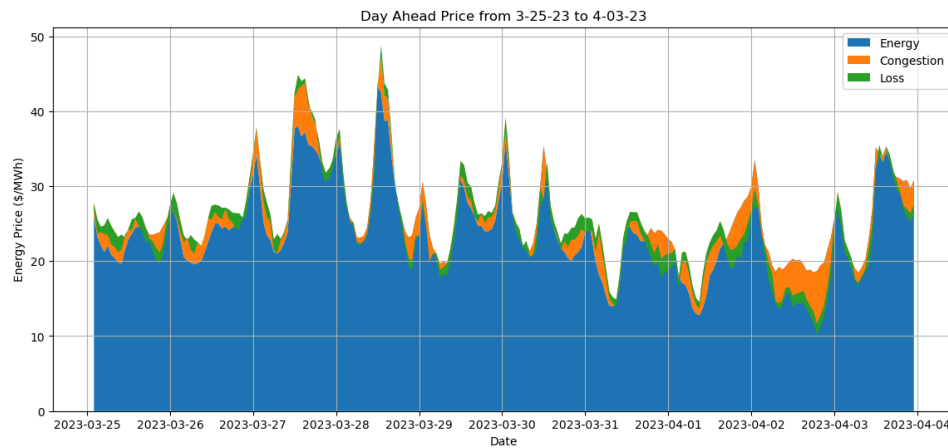
- Figures above show wind and solar energy generation of the course of a week



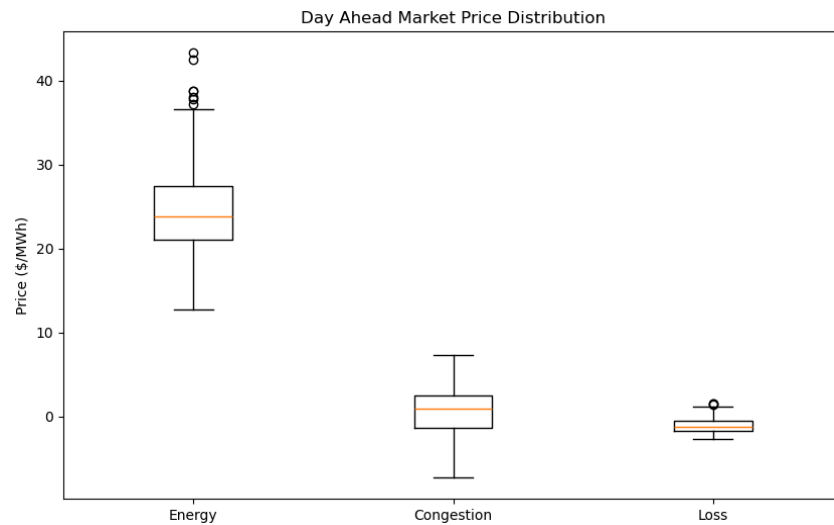
- Figure above shows Market price from the Real Time Market dataset



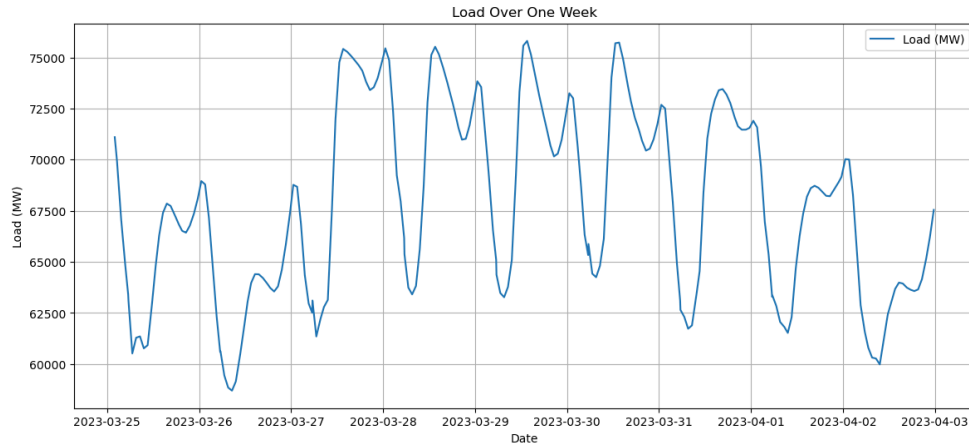
- Figure above shows the distribution of data for the Real Time Market dataset. The outliers here are rather extensive and will be something to look at in the future. This is not unexpected however since real time energy prices are difficult to predict and can vary quite a bit.



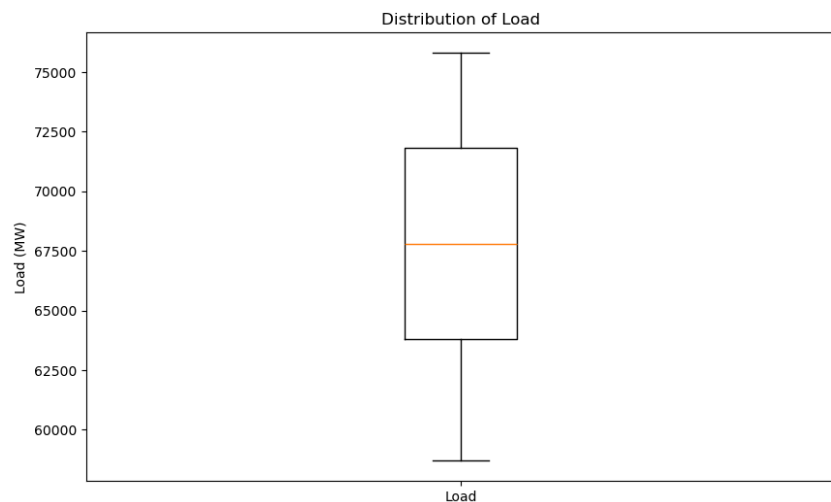
- Figure above shows the Day Ahead Price over the course of about a week. Notice how it does not match the real time price as the real prices will rarely be the same as the price set 24 hours prior.



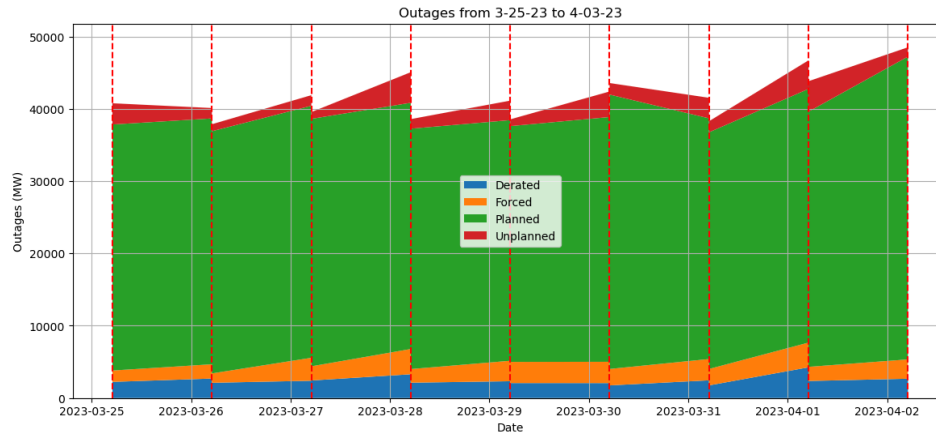
- Figure above shows the distribution of data for the Day Ahead Market with it having pretty stable data and does not have much for outliers.



- Figure above shows the energy load (MW) over the course of a week. The shape of the data makes sense with the spikes increasing when people return from work and then decreasing at night.



- Figure above shows the distribution for load and the boxplot is almost perfect. Almost no skew in data and incredibly consistent overall. (Granted this is just for the span of a week)



- Figure above shows the outages over the course of about a week. This is something to note for future project considerations since the outages are formatted in a slightly different manner, it doesn't appear to be continuous so there will have to be some data adjustments moving forward when this data gets implemented into our models.

Tommy McQuiston did the data visualization portion.

Clearly document the meaning of important columns/features and resolve ambiguous fields where possible.

The rtm (real time market) dataset contains 'Imp' (locational marginal price) which is the variable that will be the main focus of prediction for the majority of the semester. Two more variables that are important to introduce are interval_start_UTC and interval_end_UTC which represent the start and end times of each interval respectively. These two columns are found across all tables that were pulled from grid status, so they will be how we join data across much of the tables. However, one important thing to note is that the granularity of the time series data differs across tables, in the rtm table data is recorded every 5 minutes.

The dam (day ahead market) dataset also contains a 'Imp' column. This Imp however, is the forecasted Imp for each hour which is provided the day before and represents the forecasted market equilibrium of supply and demand. It is made up of 3 components, also contained and forecasted in this dataset: energy (the raw energy price), congestion (the cost of high traffic energy trying to get to the same place), and loss (the price of transporting energy over large distances). Of the components that make up the dam Imp, congestion will likely be one of the most interesting to explore how it relates to,

especially the unexpected spikes in real time Imp prices. The granularity for this table is hourly.

The outage dataset is potentially the most confusing dataset. Every day the amount of MW outages are forecasted for the highest demand hour of the next day. So the granularity of this data is the worst hour of each day. These outages are broken up into 4 categories derated_outages_mw (represent generators not operating at full capacity), unplanned_outages_mw (short notice outages due to unplanned maintenance, etc.), forced_outages_mw (sudden forced losses of equipment), and planned_outages_mw (pre scheduled maintenance outages). Using this data in conjunction with the load forecast (described below) reserve threshold (amount of wiggleroom in generation) can be calculated which may be an interesting predictor to explore as we build more complex models.

The load dataset has a granularity of 5 minutes and publishes the load_forecast, as well as the interval start and end times.

The solar and wind datasets (renewEnergy) have a granularity of 1 hour and contain the forecasted mw of solar and wind generation for every hour published the day before.

Alex Krawza has led the understanding of many of the unclear datasets and features.

Identify potential target variable(s), input features, or other variables that may be useful for the planned Data Science/ML analysis.

There are different sets for the 2 different problems that we are going to solve.

- 1) What will the next 24 hours of real time data be?
 - a) Target Variable:
 - i) Real Time LMP
 - b) Input Features:
 - i) Historical real time LMP
 - ii) Day Ahead Prices
 - iii) Solar Forecast
 - iv) Wind Forecast
 - v) Outage Forecast
- 2) What is the best buy-hold-sell schedule we can build for a battery connected to Minnesota Hub
 - a) Output Variable
 - i) A 24 hour schedule of optimal buying, holding, and selling
 - b) Input Features
 - i) A 24 hour schedule of prices both real time and day ahead

Briefly summarize the major observations from the initial exploration and explain how they may guide the next stage of the project

During the evaluation of the initial models we built, the moving average and time gap models, we concluded that using only the most recent point of prices was the best way to predict the next price. Given this knowledge, we now know that our data has strong recency bias which may limit the accuracy of traditional time series models. It is likely that the problems will be best solved by time series models anyways due to the nature of the problem, but this makes it worthwhile to explore non-time series models which are maybe better at handling non-linear patterns in predictor variables which may exist in our data.

During the evaluation of the preliminary LSTM model, we saw that there were some extreme price points which confused the model and had a large margin of error. Although these outliers are not false information, they confuse the predictions for the bulk of the data points while still having a large margin of error. With the direction of Dr. Roy, we concluded that it would be better to lose predictive power of those extreme points to the benefit of the bulk of the data. As such, we will remove the outliers and move forward with the prediction models without them.

During the exploration of the outage data, we noticed that the data has large unexpected jumps. This is due to the data only being inputted at certain hours. This presents an overarching issue with our data that there are different granularities, and the granularity of primary interest is 5 minute intervals. This is luckily an easy problem to solve and we will make a one to many association between data with low granularity and data with high granularity. By this I mean that if data is inputted hourly, there will be 12 repeats of that data in the hour to predict the real time data. While most of our data is hourly, outages are daily. It is still important that we use the 5 minute granularity even among predictor variables because we use historic real time prices to predict the future real time prices, so that predictor variable has 5 minute granularity.

Zack Sickmann and Alex Krawza have built a semester schedule that determines the continuous stages of the project